Chicago Crime Data Analysis

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Crime in Chicago(very surprising)

- 396% increase in reported crime from 2019 to 2023
- Larceny
- Aggravated Assault
- Motor Vehicle Theft

The Problem

- Inefficient Policing
- Slow Response Times
- ineffective patrolling

Objectives

- Optimization

  Patrol placement and response times

- Hotspot Detection

  Identification and prediction

- Data Driven Guidance

Stakeholder Benefits

- Law Enforcement Agencies
  - Improved Response Times
  - Enhanced Crime Deterrence
  - Mathematical Approach
- City Government
  - Allows for a strategic approach to public safety and resource allocation
- General Public
  - Safer Streets
  - Faster Emergency Care in life threatening situations

Dataset Description

- 1,456,715 instances
- 22 attributes
- Covers the years 2012 to 2017
- Most common crimes in our dataset: Theft, battery, criminal damage, narcotics, assault, other offenses
- Identified by location, year, type, case number, etc.

Preprocessing

- Dropped irrelevant columns (I.E. Case number)
- Dropped rows without coordinates
- Filled categorical missing values with ‘Unknown’
  - Preserves the row while filling the missing value
- Decomposed the Date/Time feature
  - Parsed the date string into the hour, day of the week, weekend, and month individually.
  - Added new columns in for predictive modeling
- Converted categorical variables to binary boolean values where applicable (I.E. Arrest)
- Z-Score normalization
- Took a random sample of 100,000 entries to reduce computational load
  - Consolidated to the top 10 crimes + other

Data Mining Techniques Used

Clustering:

- K-Means
  - Partition Chicago into distinct patrol zones
  - Centroids = recommended patrol points
  - Searched k = 10 -> k = 30
- DBSCAN
  - Discover crime hotspots without predefining k
- K-Fold evaluation
  - Silhouette scores

Predictive Model:

- Random Forest
  - Predicts whether a location is high-risk (Over median crime density)
  - Features: Lat/Lon, Hour, Day, Month, Weekend, arrest, and domestic
  - Feature importance: Lat then lon, followed by date and time features.

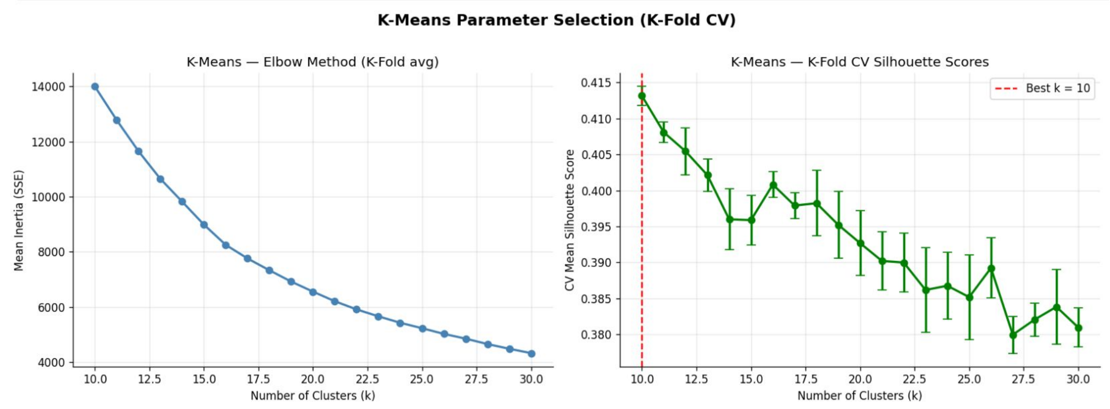
Our K-means Results

K-means clustering results:

- Searched K=10 -> K=30.
- K=10 returned the highest silhouette score.

Patrol point results:

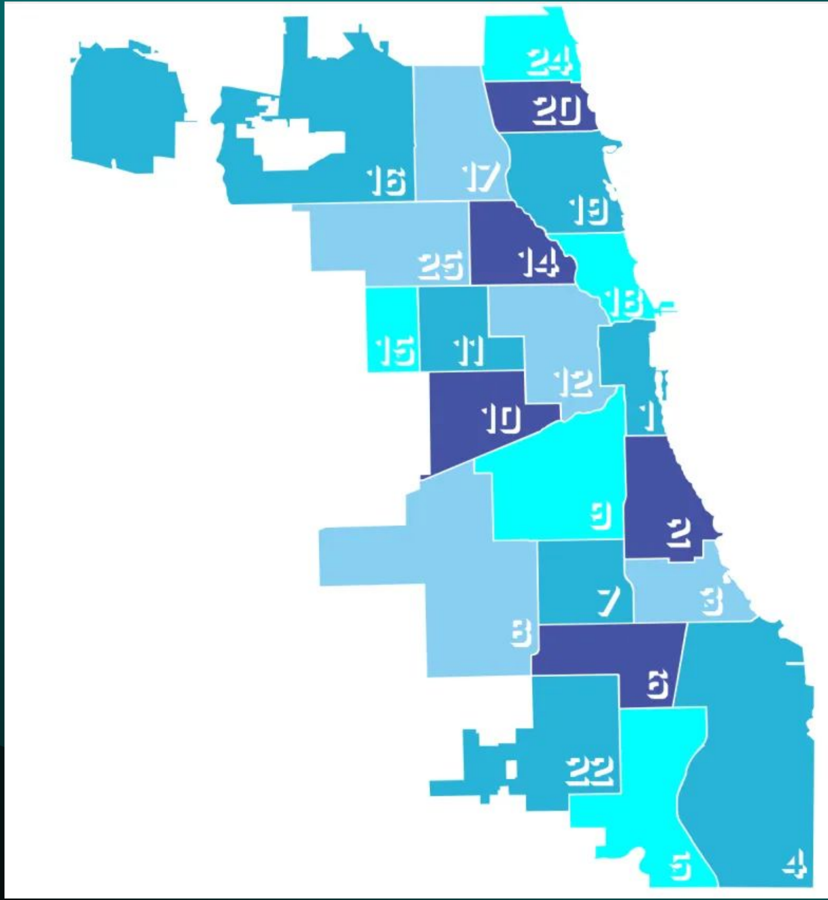
- Centroids for each cluster are our patrol points.
- Results in 10 recommended patrol points.

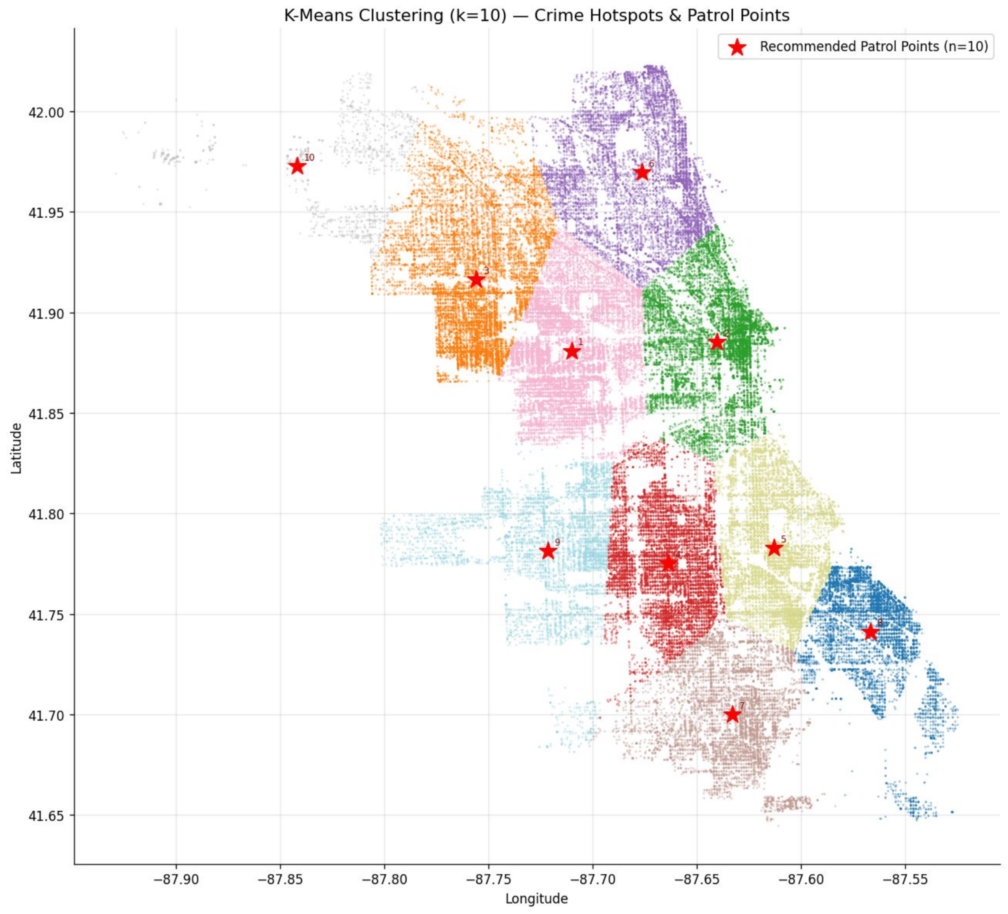


Recommended patrol points (10 zones):

Patrol Zone	Latitude	Longitude	Crime Count in Zone
6	41.880781	-87.710089	16455
2	41.885108	-87.640232	13668
1	41.916411	-87.755788	12909
3	41.775242	-87.663802	12106
8	41.782805	-87.612996	11125
4	41.969612	-87.676427	10468
5	41.700064	-87.632833	8172
0	41.741119	-87.566485	7755
9	41.781370	-87.721363	5829
7	41.972603	-87.841787	1513

Our K-means Results





Our DBSCAN Results

DBSCAN clustering results:

- Returned 57 clusters and 6464 noise points.
- Eps = 0.05, min_samples = 30 parameters returned the highest silhouette score.

Patrol point results:

- 57 patrol points returned.
- Computed geographical average of all 57 clusters found to place patrol points.
- Only created patrol points for dense crime areas, the scattered outlier crimes are treated as noise.

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DBSCAN K-Fold CV parameter grid search (5 folds):
  eps=0.05, min_s=10 | mean_sil=0.2131 ± 0.0556
  eps=0.05, min_s=20 | mean_sil=0.6098 ± 0.0812
  eps=0.05, min_s=30 | mean_sil=0.8175 ± 0.0777
  eps=0.1, min_s=10 | mean_sil=-0.3306 ± 0.0662
  eps=0.1, min_s=20 | mean_sil=-0.1890 ± 0.1418
  eps=0.1, min_s=30 | mean_sil=0.3118 ± 0.0333
  eps=0.15, min_s=10 | mean_sil=-0.0559 ± 0.0438
  eps=0.15, min_s=20 | mean_sil=0.0653 ± 0.0817
  eps=0.15, min_s=30 | mean_sil=-0.2641 ± 0.0000
  eps=0.2, min_s=10 | mean_sil=0.2446 ± 0.0997
  eps=0.2, min_s=20 | mean_sil=0.1461 ± 0.0690
  eps=0.2, min_s=30 | mean_sil=0.2946 ± 0.1756

Best DBSCAN params: eps=0.05, min_samples=30, CV silhouette=0.8175
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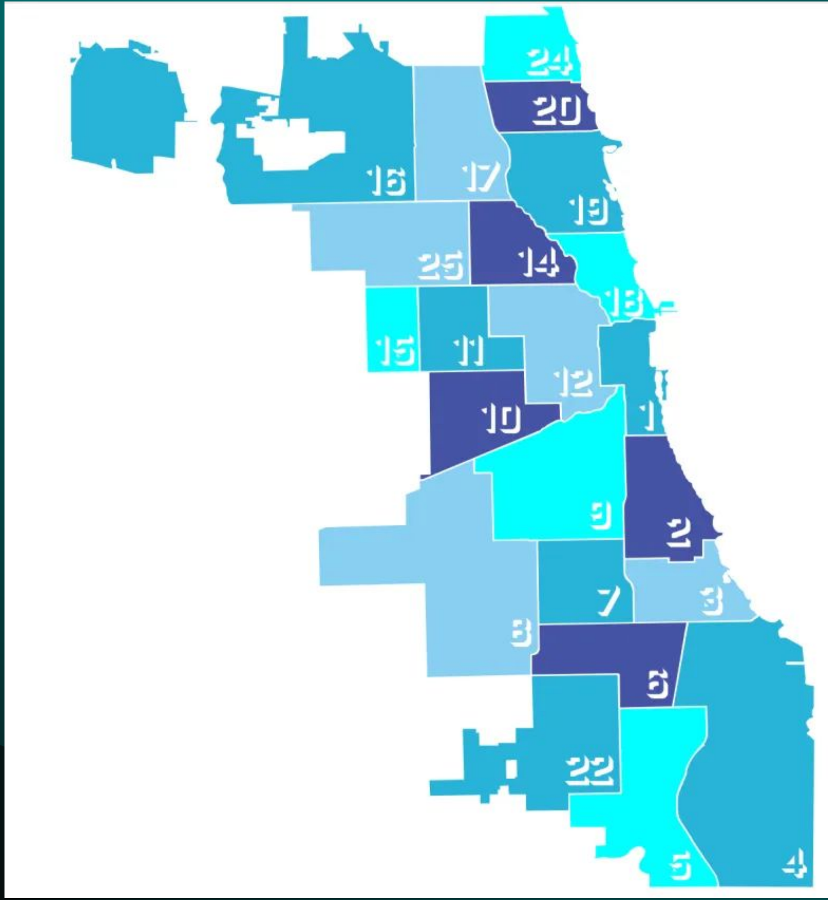
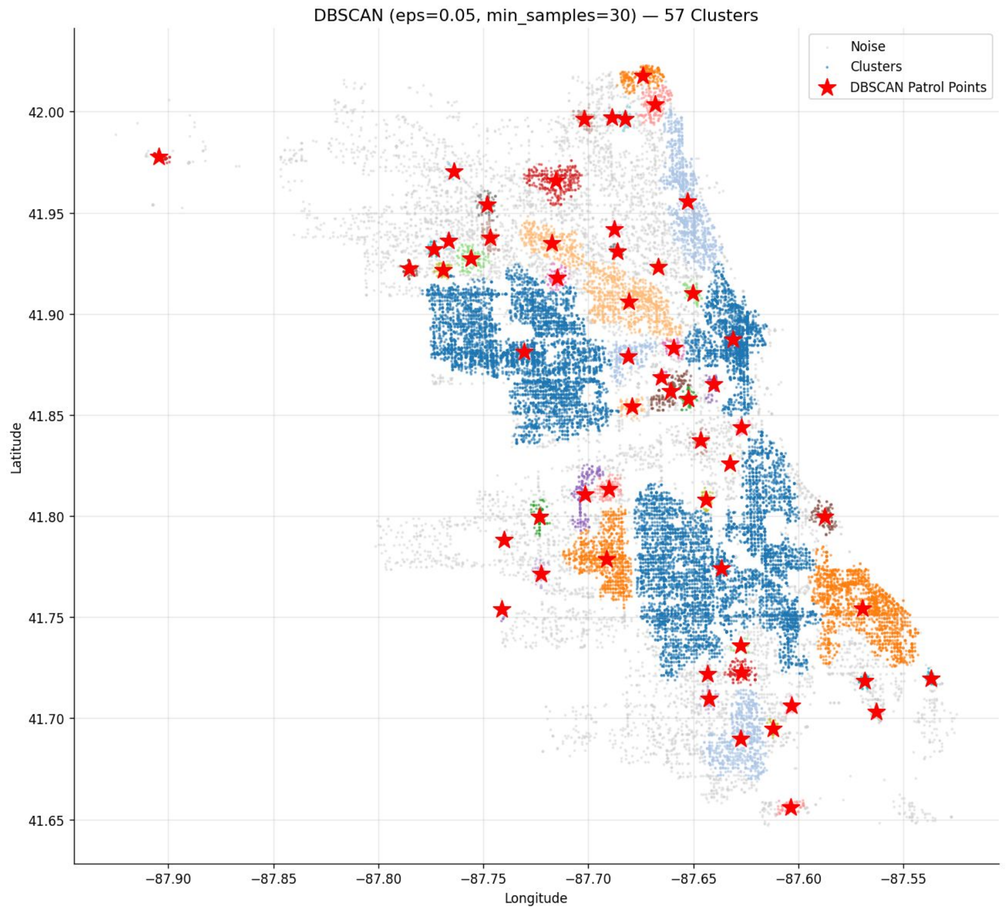
DBSCAN clusters found: 57 | Noise points: 6464 (21.5%)

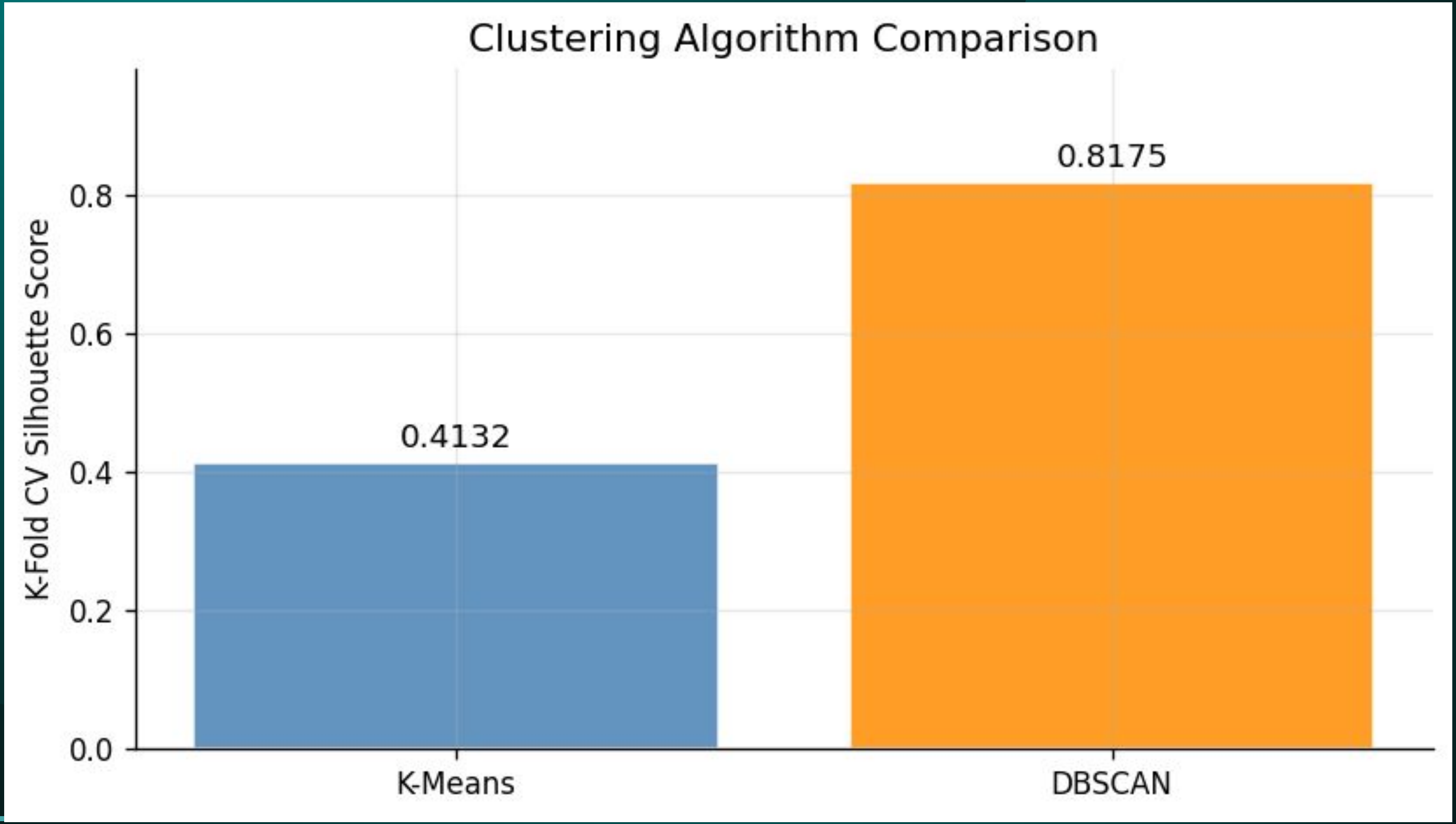
Top 15 DBSCAN patrol zones by crime density:
 Cluster  Latitude  Longitude  Crime Count
       1 41.774201 -87.636484         5892
       0 41.881166 -87.730649         5562
       2 41.887190 -87.631185         2256
       6 41.754168 -87.569628         1822
       3 41.955606 -87.652937         1201
       9 41.906010 -87.680507          904
       7 41.778813 -87.691392          883
       4 41.690056 -87.627349          746
      11 41.934803 -87.717317          340
      17 41.965660 -87.715546          308
       5 41.879001 -87.680937          270
       8 42.017535 -87.674231          239
      20 42.003573 -87.668199          178
      23 41.810732 -87.701744          175
      29 41.861753 -87.660786          169
```

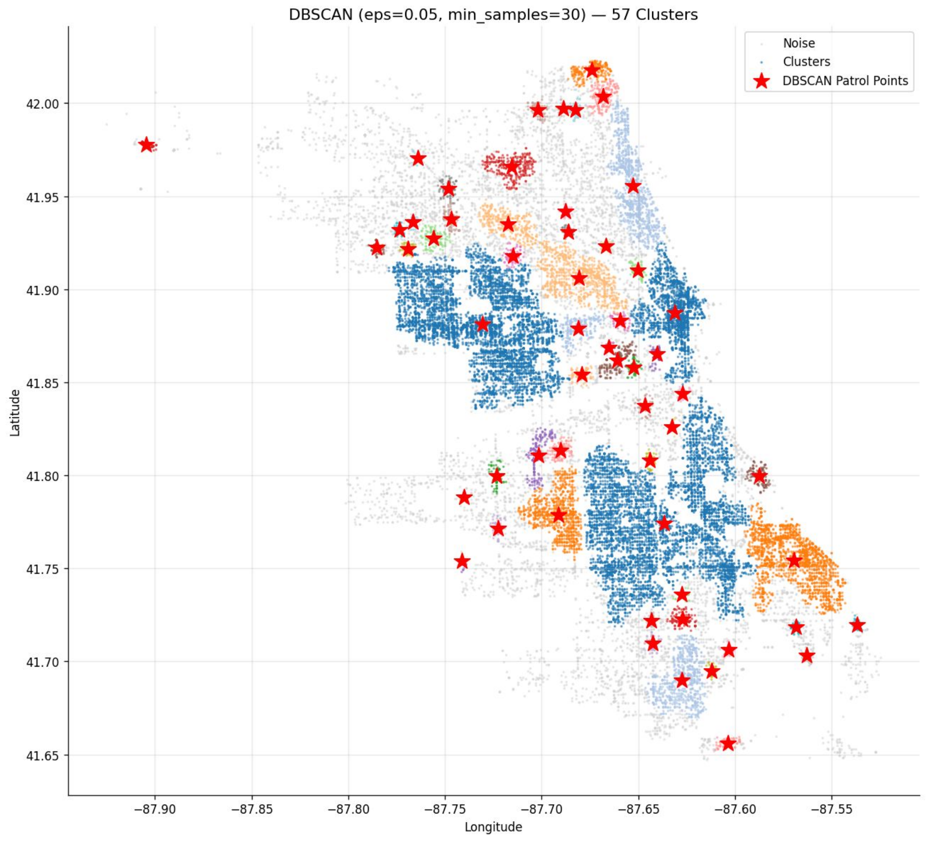
Our DBSCAN Results

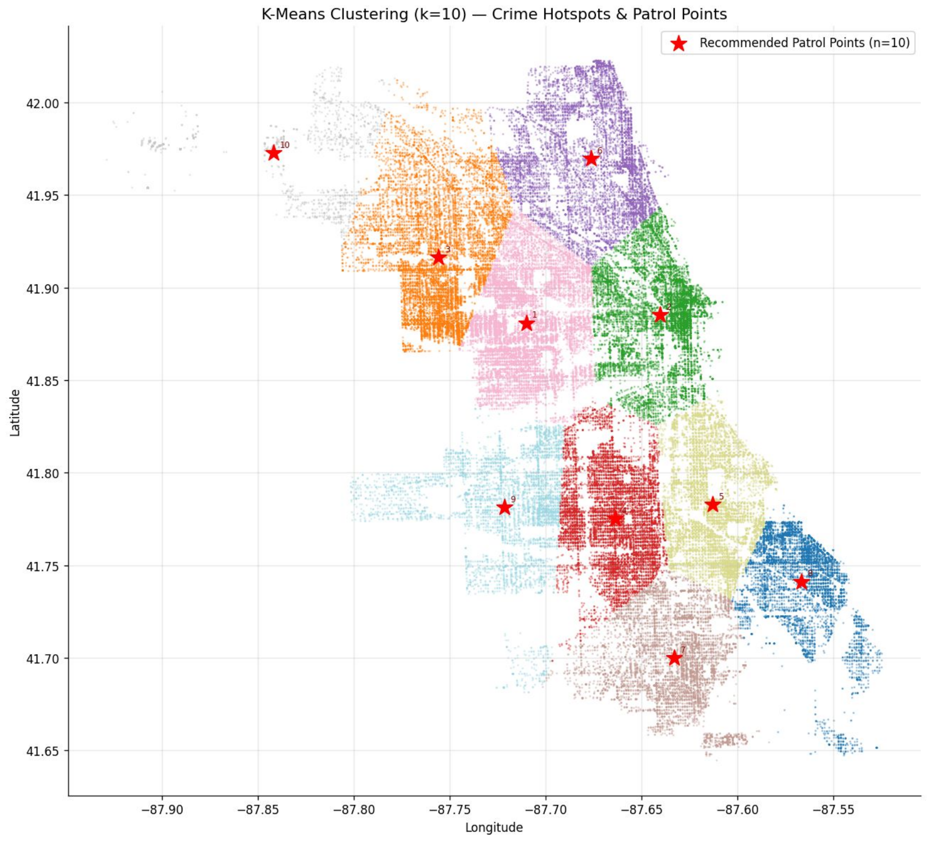




Direct head-to-head comparison of K-Means vs DBSCAN using their best K-Fold CV silhouette scores.




DBSCAN (eps=0.05, min_samples=30) — 57 Clusters
Noise
Clusters
DBSCAN Patrol Points
Latitude
Longitude
42.00
41.95
41.90
41.85
41.80
41.75
41.70
41.65
−87.90
−87.85
−87.80
−87.75
−87.70
−87.65
−87.60
−87.55


K-Means Clustering (k=10) — Crime Hotspots & Patrol Points
Recommended Patrol Points (n=10)
Latitude
Longitude
42.00
41.95
41.90
41.85
41.80
41.75
41.70
41.65
−87.90
−87.85
−87.80
−87.75
−87.70
−87.65
−87.60
−87.55

Random Forest Prediction Results

High-Risk crime zone prediction:

- Trained model to predict if k-means cluster is a high risk zone
- Rather than finding optimal patrol car location, we predict what zones have higher crime rates.
- This combined with other results, can help us determine priority spots for patrol vehicles.

Results:

- For low risk zones, it was correct 98% of the time
- For high risk zones, it was correct 99% of the time
- Precision and recall are both high meaning it’s correctly distinguishing high risk from low risk.

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High-risk zones: 5 / 10
High-risk samples: 66,012 / 98,000
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High-Risk Zone Prediction — Test Accuracy: 0.9870

	precision	recall	f1-score	support
Low Risk	0.98	0.98	0.98	6398
High Risk	0.99	0.99	0.99	13202
accuracy			0.99	19600
macro avg	0.99	0.99	0.99	19600
weighted avg	0.99	0.99	0.99	19600

Sources:

- Crime in Chicago: What You Need to Know - Council on Criminal Justice. (2025, August 26). My WordPress. https://counciloncj.org/crime-in-chicago-what-you-need-to-know/
- Crimes in Chicago. (n.d.). Www.kaggle.com. https://www.kaggle.com/datasets/currie32/crimes-in-chicago?select=Chicago_Crimes_2012_to_2017.csv
- https://chicagoreader.com/chicago-police-district-councils/